

Attention-Driven Alaryngeal Speech Enhancement with Discrete Representation Learning and Timbre-Preserving GAN Augmentation

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ABSTRACT Enhancing alaryngeal speech is a critical challenge aimed at restoring intelligibility and naturalness in individuals who have undergone laryngectomy. This paper proposes a novel end-to-end framework that leverages discrete representation learning and attention-driven alignment to improve speech quality for alaryngeal speakers. The pipeline begins with WavLM-based feature extraction from both alaryngeal and normal speech, followed by tokenization using k-means clustering. These tokens are refined via a self-supervised speech language model trained to map pathological acoustic patterns toward normal speech representations. A Conformer-based detokenizer [15] reconstructs continuous speech features from the enhanced token sequences, forming the basis for high-quality signal restoration. To further improve perceptual quality, we introduce an attention-driven feature matcher comprising three modules: (i) a feature enhancer that suppresses residual artifacts via learnable residual scaling, (ii) a multi-head attention matcher that aligns enhanced features with k-nearest neighbors from normal speech, and (iii) a quality predictor that supervises the transformation based on intelligibility-aware scores. The entire system is trained using a multi-objective loss that integrates reconstruction, contrastive, regularization, and quality-aware terms, ensuring robust convergence across acoustic conditions. A HiFi-GAN vocoder is employed to synthesize waveform output from matched features. Additionally, we introduce a data augmentation strategy based on MaskCycleGAN-VC, enhanced with a novel timbre similarity loss to preserve speaker identity across generated samples. Extensive evaluations on Thai and English datasets demonstrate that our system substantially outperforms state-of-the-art methods. Specifically, our approach achieves up to a 43.5% reduction in Frechet Audio Distance (FAD), a 5.6% reduction in mel-cepstral distortion (MCD), and attains a speaker similarity score of 0.784. These results confirm the efficacy and generalizability of our framework for practical alaryngeal speech enhancement applications.

INDEX TERMS Alaryngeal speech enhancement, speech language modeling, attention mechanism, voice conversion, data augmentation.

I. Introduction

Alaryngeal speech enhancement seeks to restore intelligibility and naturalness for individuals who have undergone laryngectomy, a procedure that removes the natural mechanism for vocal fold vibration and significantly impairs verbal com-

munication. Traditional alternatives such as electrolarynx devices, esophageal speech, and tracheoesophageal puncture (TEP) provide substitute sound sources but often result in limited prosody, reduced clarity, and speaker variability.

These methods typically require extensive training and still fail to replicate the rich expressiveness of natural speech.

Earlier computational attempts to improve alaryngeal speech include statistical models such as Gaussian Mixture Models (GMM) [1] and formant synthesis [2], which have proven insufficient in handling the articulatory variability and spectral distortion inherent in alaryngeal phonation. More recent deep learning-based approaches such as kNN-VC [3], which combines WavLM embeddings [4], k-nearest neighbor matching, and a neural vocoder, have shown promise in improving intelligibility. However, these models primarily rely on direct spectral mapping and lack mechanisms to model broader linguistic or prosodic structures. Moreover, they often underperform in low-resource scenarios, where the scarcity of authentic alaryngeal speech samples impedes model generalization. To address this, recent works such as the Phoneme Hallucinator [5] have explored augmentation techniques for improving robustness.

In this paper, we propose a comprehensive framework that integrates language modeling, attention-based alignment, and adversarial data augmentation to substantially improve alaryngeal speech enhancement. The system architecture is depicted in Figure 1. Our method begins by extracting WavLM-based speech representations from both alaryngeal and normal speech. These are tokenized via a k -means quantizer and refined using a Speech Language Model (SpeechLM), following the SELM framework [6]. This step transforms coarse alaryngeal tokens into a more linguistically plausible sequence. A Conformer-based detokenizer reconstructs continuous feature vectors, yielding an intermediate enhanced representation.

Although SELM-based enhancement improves token-level fidelity, residual artifacts from the alaryngeal source may remain. To further refine the signal, we introduce an Attention-Driven Feature Matcher (ADFM), a three-stage alignment pipeline comprising: (i) a feature enhancer, (ii) an attention-based matcher leveraging k -nearest normal embeddings, and (iii) a quality predictor that adaptively reweights features to preserve naturalness and speaker identity. The final matched features are synthesized using a HiFi-GAN vocoder [7], producing natural-sounding speech with high timbral fidelity.

To overcome data scarcity, we adopt a novel augmentation strategy using MaskCycleGAN-VC [8], enhanced with a Timbre-Preserving Loss. Our augmentation employs structured corruption strategies—including Consecutive Time Masking (CTM) [11], Articulatory Dimension Masking (ADM) [11], and SpecAugment [12]—to synthesize diverse alaryngeal-like utterances while maintaining speaker identity.

The key contributions of this work are as follows:

- 1) **Discrete Speech Representation Learning for Alaryngeal Enhancement:** We adopted the SELM framework [6] incorporating a speech language model with K-Means tokenization and Conformer-based detokenization to refine alaryngeal speech features, trans-

forming them into enhanced representations that better resemble normal speech characteristics.

- 2) **Attention-Driven Feature Matching:** We introduce a novel alignment module that contextually matches enhanced alaryngeal features to normal speech using multi-head attention over k -nearest WavLM embeddings, guided by a trainable quality predictor.
- 3) **Multi-Mask GAN-Based Data Augmentation:** We integrate MaskCycleGAN-VC with a timbre-aware loss and diverse masking strategies to generate realistic, speaker-consistent alaryngeal speech for data augmentation in low-resource settings.

Overall, our integrated system combines the strengths of linguistic modeling, attention-based feature alignment, and speaker-aware data augmentation to significantly enhance alaryngeal speech naturalness and intelligibility across languages and conditions.

TABLE 1. Commonly Used Notations.

Notation	Description
T	Sequence length
F, D	Feature dimension (1024 for WavLM)
s	Input speech segment
$\mathbf{r}_t$	WavLM embedding at time step t
μ_k	k -means cluster centroid for cluster k
$\mathbf{x}$	Initial discrete token sequence
$\hat{\mathbf{x}}$	Refined token sequence from SELM
$\mathbf{z}_t$	Contextualized embeddings from attention
$q(\hat{y}_t \mathbf{x})$	Predicted token probability distribution
$p(y_t)$	Target token probability distribution
$\mathbf{e}_t$	Token embedding with positional encoding
$\mathbf{h}_{\text{detok}}$	Reconstructed features from detokenizer
$\mathbf{h}_{\text{enhanced}}$	Enhanced features from Feature Enhancer
$\mathbf{h}_{\text{matched}}$	Matched features from Attention Matcher
$\mathbf{h}_{\text{final}}$	Final enhanced features from ADFM
$\bar{\mathbf{h}}_{\text{neighbors}}$	Average of k -nearest neighbor features
τ	Temperature parameter for contrastive loss
$\lambda_{\text{rec}}, \lambda_{\text{reg}}, \lambda_{\text{con}}, \lambda_{\text{qual}}$	Loss weights
$\mathcal{L}_{\text{speechLM}}$	SELM speech language model loss
$\mathcal{L}_{\text{detok}}$	Detokenizer reconstruction loss
$\mathcal{L}_{\text{rec}}$	Reconstruction loss of Attention Matcher
$\mathcal{L}_{\text{reg}}$	Regularization loss of Feature Enhancer
$\mathcal{L}_{\text{con}}$	Contrastive loss of Attention Matcher
$\mathcal{L}_{\text{qual}}$	Quality loss from Quality Predictor
$\mathcal{L}_{\text{total}}$	Combined total training loss
$\hat{q}$	Quality score

This paper is organized as follows. Section II describes the discrete representation and speech feature enhancement pipeline. Section III details the Attention-Driven Feature Matcher. Section IV outlines the Multi-Mask GAN-based data augmentation with timbre preservation. Section V re-

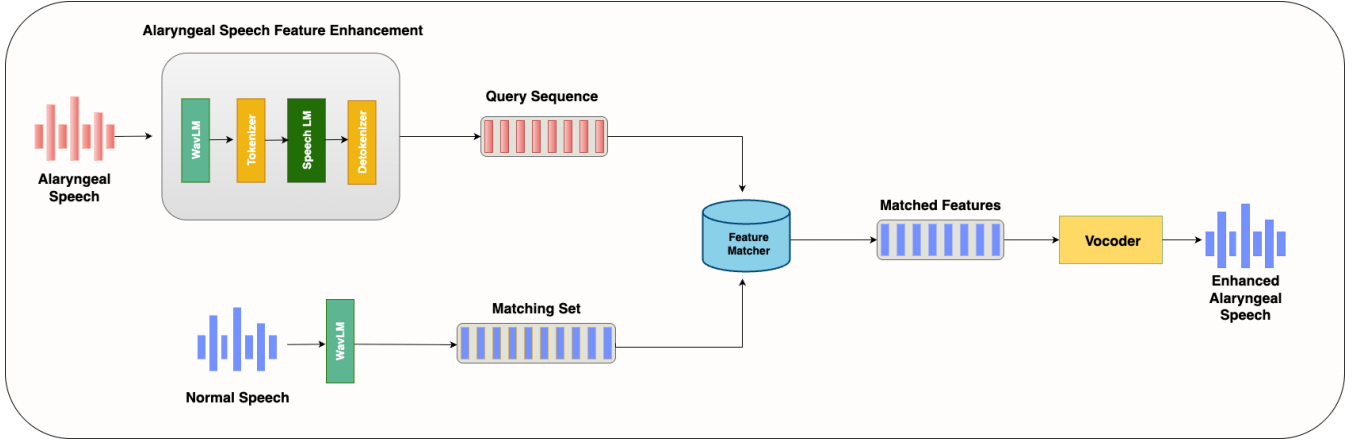


FIGURE 1. Block diagram of the proposed alaryngeal speech enhancement system.

ports on experimental setup and results. Finally, Section VI concludes the paper and discusses future work.

II. Discrete Speech Representation Learning for Alaryngeal Enhancement

Our approach adapts the SELM framework [6] for alaryngeal speech enhancement through a three-stage pipeline: (1) acoustic tokenization via WavLM and k-means clustering, (2) token refinement using SELM’s speech language model, and (3) continuous feature reconstruction via Conformer decoder. This discrete-to-continuous modeling paradigm enables structured enhancement of alaryngeal speech by operating in the symbolic token space before reconstructing high-quality acoustic features.

A. Speech Feature Tokenization

To transform raw speech signals into a structured representation, we employ WavLM, a self-supervised speech model designed to capture fine-grained high-dimensional acoustic features. Given an input speech segment s , whether alaryngeal or normal, WavLM extracts a sequence of high-dimensional embeddings:

$$\mathbf{r} = \{\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_T\}, \quad \mathbf{r}_t \in \mathbb{R}^F, \quad (1)$$

where T represents the sequence length, and $F = 1024$ is the feature dimension. These embeddings preserve both spectral and temporal speech characteristics, forming a robust latent representation.

To further structure the extracted features, we discretize the embeddings into tokenized representations using k-means clustering. The clustering process minimizes intra-cluster variance, yielding a set of cluster centroids:

$$\mu_k = \arg \min_{\mu_k} \sum_{t=1}^T \|\mathbf{r}_t - \mu_k\|^2, \quad k \in \{1, 2, \dots, K\} \quad (2)$$

Each speech frame embedding is mapped to its closest centroid, forming a sequence of discrete tokens:

$$\mathbf{x} = \{x_1, x_2, \dots, x_T\}, \quad x_t = \arg \min_k \|\mathbf{r}_t - \mu_k\| \quad (3)$$

This discretization serves as a fundamental step for token-based speech modeling, improving interpretability and making feature transformation more structured and tractable for downstream enhancement tasks. The resulting token sequence $\mathbf{x}$ serves as input to the subsequent SELM-based refinement stage.

B. Token Sequence Refinement via Speech Language Model

While vector quantization produces a concise symbolic representation, the resulting alaryngeal tokens may exhibit artifacts stemming from impaired articulation patterns. Adopting SELM’s methodology [6], we utilize a transformer-based speech language model that frames alaryngeal speech enhancement as a specialized form of mask prediction.

Following SELM’s approach, we view alaryngeal speech enhancement as mask prediction where distorted portions are treated as naturally “masked” areas requiring contextual prediction. The approach treats enhancement as token transformation where alaryngeal tokens serve as input and normal speech tokens serve as targets.

We employ SELM’s decoder-only transformer architecture with attention mechanisms applied to input tokens for comprehensive contextual modeling. The model processes alaryngeal tokens through audio embedding, positional encoding, and multi-head self-attention layers to predict corresponding normal speech tokens:

$$q(\hat{y}_t | \mathbf{x}) = \text{softmax}(W \mathbf{z}_t + b) \quad (4)$$

where $\mathbf{x}$ represents the initial alaryngeal tokens and $\mathbf{z}_t$ are contextualized embeddings from the attention layers.

The training objective follows SELM’s formulation:

$$\mathcal{L}_{\text{speechLM}} = - \sum_{t=1}^{T'} p(y_t) \cdot \log \left(\frac{q(\hat{y}_t)}{p(y_t)} \right) \quad (5)$$

This enables direct transformation from alaryngeal to normal speech patterns through contextual understanding of naturally corrupted tokens. The refined token sequence $\hat{\mathbf{x}} = \{\hat{x}_1, \hat{x}_2, \dots, \hat{x}_{T'}\}$ is then passed to the acoustic reconstruction stage.

C. Acoustic Feature Reconstruction via Conformer Decoder

Following token refinement, the symbolic sequence $\hat{\mathbf{x}}$ must be mapped back to high-resolution acoustic features suitable for downstream synthesis or evaluation. To this end, we employ a Conformer-based decoder, chosen for its ability to effectively model both long-range dependencies and local temporal structures critical for speech.

Each refined token $\hat{x}_t$ is first transformed into a continuous embedding vector and enriched with positional information:

$$\mathbf{e}_t = \text{Embed}(\hat{x}_t) + \text{PosEnc}(t), \quad (6)$$

where $\text{Embed}(\cdot)$ denotes a learned embedding function and $\text{PosEnc}(t)$ provides sinusoidal position encodings to retain temporal order.

The sequence $\{\mathbf{e}_t\}$ is then fed into a multi-layer Conformer decoder, which integrates multi-head self-attention for global context modeling and depthwise separable convolutions for local feature refinement. The decoder outputs a sequence of reconstructed frame-level features:

$$\mathbf{h}_{\text{detok}} = \{\mathbf{h}_{\text{detok},1}, \dots, \mathbf{h}_{\text{detok},T'}\}, \quad \mathbf{h}_{\text{detok},t} \in \mathbb{R}^{1024}. \quad (7)$$

To ensure fidelity to natural speech, the decoder is trained with a reconstruction loss that minimizes the mean squared error (MSE) between the predicted features and ground-truth WavLM embeddings from normal speech:

$$\mathcal{L}_{\text{detok}} = \frac{1}{T'} \sum_{t=1}^{T'} \|\mathbf{h}_{\text{target},t} - \mathbf{h}_{\text{detok},t}\|^2. \quad (8)$$

The reconstructed features $\mathbf{h}_{\text{detok}}$ serve as enhanced speech representations that are subsequently processed by the attention-driven feature matcher for further quality improvement.

D. End-to-End Training Objective

To jointly optimize the token transformation and acoustic reconstruction capabilities of our model, we adopt a unified end-to-end training strategy. The total loss function integrates both SELM's speech language modeling and frame-level acoustic regression:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{speechLM}} + \lambda \mathcal{L}_{\text{detok}}, \quad (9)$$

where $\mathcal{L}_{\text{speechLM}}$ denotes the SELM token transformation loss, $\mathcal{L}_{\text{detok}}$ represents the acoustic feature reconstruction loss, and λ is a weighting coefficient that balances the importance of token-level enhancement against acoustic fidelity.

This joint objective enables the model to learn the transformation from alaryngeal to normal speech patterns in the discrete token space while simultaneously ensuring that the

reconstructed speech features are perceptually and acoustically aligned with natural voice characteristics. The discrete-to-continuous design is particularly beneficial for alaryngeal speech, as it addresses both phonetic inconsistencies and spectral distortions commonly observed in pathological speech. By bridging SELM's conceptual masking approach with continuous acoustic modeling, the proposed framework supports intelligible and natural-sounding speech enhancement even under challenging alaryngeal input conditions.

III. Attention-Driven Feature Matcher

A. Feature Enhancer

The output features from the Conformer-based detokenizer, while reconstructed from the refined tokens, may still contain residual artifacts from the alaryngeal speech input. To further improve the quality of these features and facilitate better alignment with normal speech representations, we introduce a Feature Enhancement Network containing feature enhancer, attention matcher and quality predictor as shown in Figure 3.

The Feature Enhancer aims to suppress remaining artifacts while amplifying informative content in the high-dimensional feature space (1024-dimensional WavLM embeddings). Given that direct transformation in such high-dimensional spaces can lead to training instability, we design a lightweight residual feedforward network that processes each feature vector individually:

$$\mathbf{h}_{\text{enhanced}} = \alpha \cdot f_{\text{MLP}}(\mathbf{h}_{\text{detok}}) + (1 - \alpha) \cdot \mathbf{h}_{\text{detok}} \quad (10)$$

where $\mathbf{h}_{\text{detok}} \in \mathbb{R}^{1024}$ represents the input features from the Conformer detokenizer, f_{MLP} denotes the multi-layer perceptron transformation, and α is a learnable parameter that controls the balance between enhanced and original features.

The MLP architecture follows a bottleneck design with three linear layers. Each linear transformation is followed by layer normalization, ReLU activation, and dropout regularization with a dropout to prevent overfitting.

The residual connection preserves the original representations while allowing the network to learn incremental improvements. The enhanced features $\mathbf{h}_{\text{enhanced}}$ serve as refined representations of alaryngeal speech that are subsequently fed to the attention matcher module for alignment with normal speech patterns.

To prevent the enhanced features from deviating excessively from the original alaryngeal characteristics while allowing meaningful improvements, we employ a regularization loss:

$$\mathcal{L}_{\text{reg}} = \frac{1}{D} \sum_{i=1}^D (h_{\text{enhanced},i} - h_{\text{alaryngeal},i})^2 \quad (11)$$

This loss ensures that the enhancement process maintains the essential characteristics of the input while suppressing artifacts.

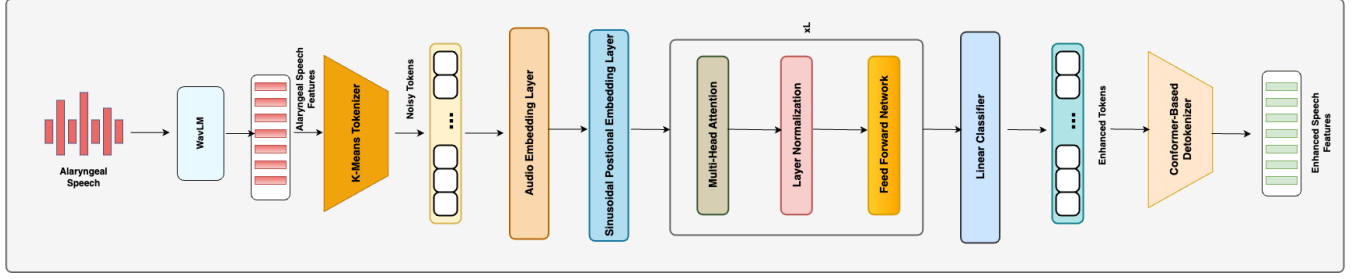


FIGURE 2. Alaryngeal Speech Feature Enhancement.

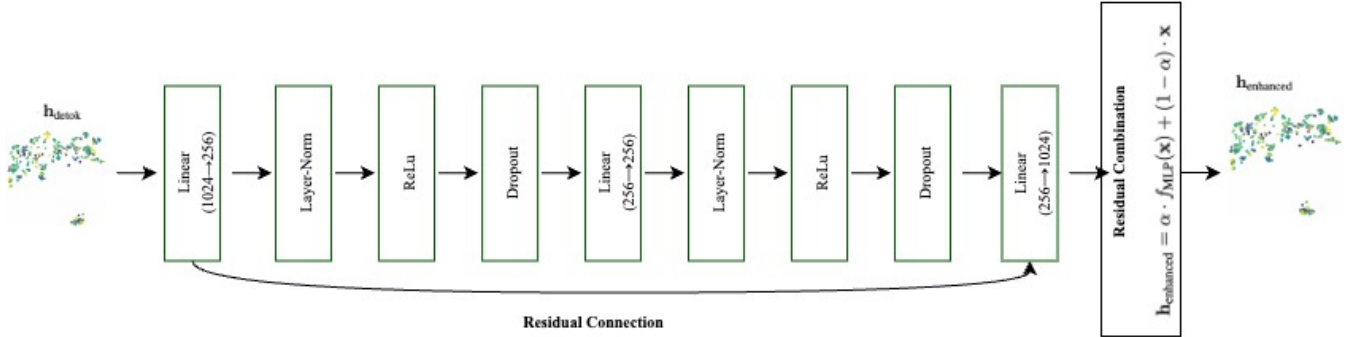


FIGURE 3. Feature Enhancer

B. Attention Matcher

The enhanced features $\mathbf{h}_{\text{enhanced}}$ from the Feature Enhancer serve as input to the Attention Matcher as shown in Figure 4, which aligns these refined alaryngeal features with a set of normal speech candidates using a multi-head attention mechanism with 4 attention heads.

For each enhanced feature vector $\mathbf{h}_{\text{enhanced}} \in \mathbb{R}^{1024}$, we first retrieve the top- k (where $k = 10$) most similar normal speech features using cosine-based k -Nearest Neighbors search. To improve training efficiency, these KNN matches are pre-computed during dataset initialization using GPU acceleration for faster computation. The candidate features serve as keys and values in the attention mechanism:

$$\mathbf{K}, \mathbf{V} = \{\mathbf{h}_{\text{normal},1}, \mathbf{h}_{\text{normal},2}, \dots, \mathbf{h}_{\text{normal},k}\} \in \mathbb{R}^{k \times 1024} \quad (12)$$

where each $\mathbf{h}_{\text{normal},i}$ represents a normal speech feature vector selected based on cosine similarity to $\mathbf{h}_{\text{enhanced}}$.

The multi-head attention mechanism computes a weighted combination of the candidate normal features:

$$\mathbf{h}_{\text{matched}}, \mathbf{w} = \text{MultiHeadAttention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) \quad (13)$$

where $\mathbf{Q} = \mathbf{h}_{\text{enhanced}}$ (expanded to sequence dimension), and $\mathbf{w} \in \mathbb{R}^k$ represents the attention weights. The attention output is then processed through residual connections, layer normalization, and a feedforward projection:

$$\mathbf{h}_{\text{attn}} = \text{LayerNorm}(\mathbf{h}_{\text{matched}} + \mathbf{h}_{\text{enhanced}}) \quad (14)$$

$$\mathbf{h}_{\text{final}} = \text{LayerNorm}(\text{FFN}(\mathbf{h}_{\text{attn}}) + \mathbf{h}_{\text{attn}}) \quad (15)$$

where FFN denotes a feedforward network with hidden dimension 256. This design enables refined alignment while preserving the enhanced features through residual connections and maintains interpretability through attention weights.

The attention matcher is optimized using two complementary loss functions. First, a reconstruction loss encourages the matched features to align with the average of the selected normal speech neighbors:

$$\mathcal{L}_{\text{rec}} = \frac{1}{D} \sum_{i=1}^D (h_{\text{matched},i} - \bar{h}_{\text{neighbors},i})^2 \quad (16)$$

where $\bar{h}_{\text{neighbors}} = \frac{1}{k} \sum_{j=1}^k \mathbf{h}_{\text{neighbor},j}$ is the average of k nearest neighbor features.

A contrastive loss with temperature $\tau = 0.1$ ensures matched features are closer to targets than random samples:

$$\mathcal{L}_{\text{con}} = -\log \left(\frac{\exp(s_{\text{pos}}/\tau)}{\exp(s_{\text{pos}}/\tau) + \exp(s_{\text{neg}}/\tau)} \right) \quad (17)$$

where $s_{\text{pos}} = \text{sim}(\mathbf{h}_{\text{matched}}, \bar{\mathbf{h}}_{\text{neighbors}})$ and $s_{\text{neg}} = \text{sim}(\mathbf{h}_{\text{matched}}, \mathbf{h}_{\text{random}})$.

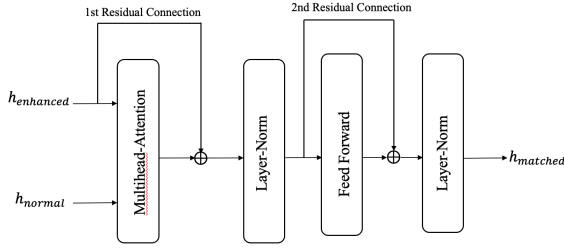


FIGURE 4. Attention Matcher.

C. Quality Predictor

The Quality Predictor evaluates the perceptual quality of the transformed features. It consists of a two-layer MLP with ReLU and Sigmoid activations, predicting a scalar quality score $\hat{q} \in [0, 1]$ for each sample:

$$\hat{q} = \sigma(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \cdot \mathbf{h}_{\text{final}} + \mathbf{b}_1) + \mathbf{b}_2) \quad (18)$$

where $\mathbf{W}_1 \in \mathbb{R}^{128 \times 1024}$, $\mathbf{W}_2 \in \mathbb{R}^{1 \times 128}$, and σ is the sigmoid function. A higher score indicates higher similarity to natural speech.

The quality predictor is trained using a quality loss that encourages the model to predict high quality scores for well-transformed features:

$$\mathcal{L}_{\text{qual}} = (\hat{q} - 1)^2 \quad (19)$$

This encourages the model to predict high quality scores for well-enhanced features. The overall training objective combines all loss components:

$$\begin{aligned} \mathcal{L}_{\text{total}} = & \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}} + \lambda_{\text{con}} \mathcal{L}_{\text{con}} \\ & + \lambda_{\text{qual}} \mathcal{L}_{\text{qual}} \end{aligned} \quad (20)$$

The combined loss function balances multiple training objectives. The reconstruction loss ($\mathcal{L}_{\text{rec}}$) receives the highest weight to ensure proper alignment with normal speech, while regularization ($\mathcal{L}_{\text{reg}}$) prevents excessive deviation from the original features. The contrastive component ($\mathcal{L}_{\text{con}}$) improves feature discrimination, and quality assessment ($\mathcal{L}_{\text{qual}}$) provides additional perceptual guidance. This design produces natural-sounding speech while preserving speaker characteristics.

IV. Multi-Mask GAN-Based Data Augmentation with Timbre-Preserving Supervision

Alaryngeal speech data is extremely scarce due to the specialized clinical nature of laryngectomy and the logistical challenges in collecting such speech samples. This scarcity presents a bottleneck in training data-intensive models such as voice conversion (VC) systems. To address this, we propose a novel data augmentation pipeline that synthesizes

realistic alaryngeal speech by leveraging a multi-mask generative adversarial network (GAN) architecture integrated with timbre-preserving supervision.

Our framework builds upon MaskCycleGAN-VC, which extends the classic CycleGAN-VC [17] by introducing structured spectrotemporal perturbations during training. We further enhance the model by (1) designing a multi-mask augmentation strategy that exploits diverse spectrogram masking operations and (2) integrating a speaker-aware loss that explicitly preserves timbre during generation.

A. Spectrogram Masking for Robust Data Augmentation

To improve generalization and robustness in training the GAN-based speech enhancement system, we apply spectrogram-level masking techniques that simulate realistic degradations in the acoustic input. The Mel-spectrogram is used as the input representation due to its compactness and strong correlation with human auditory perception.

Let $\mathbf{M} \in \mathbb{R}^{B \times L}$ denote the Mel-spectrogram of a speech segment, where B is the number of Mel frequency bins and L is the number of time frames. We introduce a set of stochastic masking operations that follow the *Filling-In-Frame* (FIF) principle—deliberately removing structured portions of the input to encourage the generator to infer plausible completions. These masks are applied dynamically during training to introduce diversity and enforce invariance to temporal and spectral perturbations.

The masking strategies include:

- **Consecutive Time Masking (CTM):** Randomly removes contiguous time spans from the spectrogram, i.e., $\mathbf{M}_{:,l_s:l_e} = 0$, where $[l_s, l_e]$ is a randomly chosen interval. This forces the generator to extrapolate temporally, promoting continuity reconstruction across missing speech segments.
- **Articulatory Dimension Masking (ADM):** Zeros out entire frequency regions, $\mathbf{M}_{b_s:b_e,:} = 0$, simulating articulatory suppression such as loss of formant or pitch information. This encourages the model to reconstruct missing phonetic and prosodic details from contextual cues.
- **SpecAugment** [13]: A widely used augmentation technique that combines both time and frequency masking. Originally proposed for ASR, it has demonstrated strong regularization effects in various speech modeling tasks. We adopt it here as a robust baseline for comparison.

B. Timbre Similarity Loss for Speaker Identity Preservation

Preserving speaker identity is a critical requirement in voice conversion systems, particularly in clinical and rehabilitative contexts where the listener's familiarity with the speaker's voice facilitates effective communication. To enforce timbral

consistency in the synthesized speech, we introduce a timbre similarity loss based on speaker embeddings.

We leverage the Resemblyzer model [18], a state-of-the-art embedding extractor trained for speaker discrimination. Let v_s and v_a denote the speaker embeddings corresponding to the synthesized and reference alaryngeal utterances, respectively. We compute their cosine similarity as:

$$T_{\text{sim}}(v_s, v_a) = \frac{v_s^\top v_a}{\|v_s\| \cdot \|v_a\|} \quad (21)$$

To encourage the generator to retain speaker-specific vocal attributes, we define the timbre similarity loss as:

$$\mathcal{L}_{\text{te}} = 1 - T_{\text{sim}}(v_s, v_a) \quad (22)$$

This loss acts as a perceptually informed auxiliary constraint, complementing the primary spectral reconstruction objectives. By operating in a high-level embedding space, it ensures that speaker identity is preserved even when low-level acoustic features are perturbed or restructured. Empirically, we observe that incorporating $\mathcal{L}_{\text{te}}$ improves listener-rated voice naturalness and speaker recognizability without compromising intelligibility.

C. GAN Optimization with Multi-Objective Loss

To unify the proposed modules into a cohesive generative framework, we extend the standard MaskCycleGAN-VC objective to incorporate speaker-aware and perceptually grounded constraints. Our multi-objective loss balances adversarial realism, structural consistency, and speaker identity preservation to ensure high-quality and personalized alaryngeal speech synthesis.

The overall training objective is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{adv}} + \lambda_{\text{cyc}} \mathcal{L}_{\text{cyc}} + \lambda_{\text{id}} \mathcal{L}_{\text{id}} + \mathcal{L}_{\text{adv}2} + \lambda_{\text{te}} \mathcal{L}_{\text{te}}, \quad (23)$$

where each component serves a distinct role:

- $\mathcal{L}_{\text{adv}}$: Primary adversarial loss that encourages the generator to produce acoustically realistic speech samples.
- $\mathcal{L}_{\text{cyc}}$: Cycle-consistency loss ensuring reversible translation between normal and alaryngeal speech domains, thereby enforcing structural integrity.
- $\mathcal{L}_{\text{id}}$: Identity loss that preserves content and speaker characteristics under self-mapping conditions.
- $\mathcal{L}_{\text{adv}2}$: Auxiliary adversarial loss used to stabilize GAN training by introducing a second discriminator stream.
- $\mathcal{L}_{\text{te}}$: Timbre similarity loss (as described in Section B) that maintains speaker identity in the generated speech.

The trade-off parameters λ_{cyc} , λ_{id} , and λ_{te} are empirically tuned to balance these competing objectives, ensuring both intelligibility and speaker recognizability in the output.

The proposed framework synergistically combines structured spectrogram corruption with adversarial learning and embedding-level timbre supervision. This enables the generator to generalize effectively under diverse distortions and limited training data. Crucially, the inclusion of $\mathcal{L}_{\text{te}}$

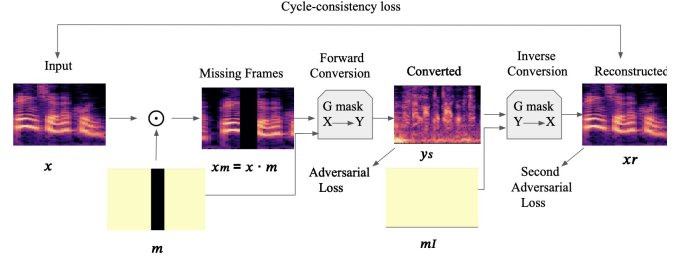


FIGURE 5. Pipeline of Filling-In-Frame (FIF) utilizing Consecutive Time Masking (CTM).

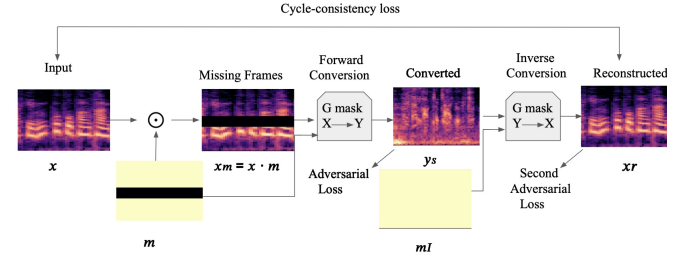


FIGURE 6. Pipeline of FIF utilizing Articulatory Dimension Masking (ADM).

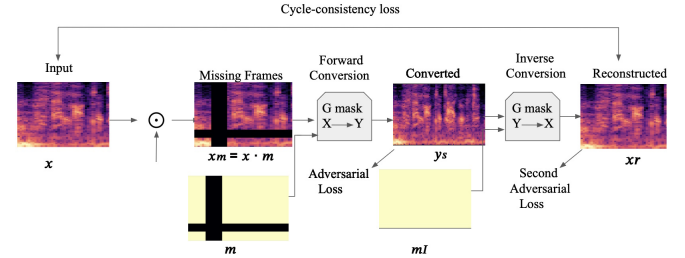


FIGURE 7. Pipeline of FIF utilizing SpecAugment.

ensures that voice identity is preserved, a key requirement in clinical deployments where familiarity and personalization significantly impact communicative effectiveness. Our results suggest that this integrated strategy not only enhances speech quality but also improves listener recognition and subjective acceptability in alaryngeal speech applications.

V. Experiment

A. Component-wise Ablation Study

We conducted ablation studies to analyze the individual contributions of each component in our proposed workflow using 50 test samples from 3 different alaryngeal speakers. With kNN-VC as our baseline model, we systematically evaluated the impact of integrating the Speech Language Model and Attention-Driven Feature Matcher through objective metric comparison using Frechet Audio Distance. The baseline performance and component-wise improvements are presented in Figure 8.

1) Impact of Speech Language Model on Source Enhancement

To assess the efficacy of the Speech Language Model, we inserted it upstream of the kNN-VC pipeline to enhance input features while keeping all other components unchanged. This configuration simulates the effect of preprocessing the source speech before feature matching.

As illustrated in Figure 8, incorporating SLM reduced the FAD from 7.518 (baseline) to 6.563, reflecting a relative improvement of approximately 12.7%. This demonstrates that denoising and enhancing alaryngeal speech prior to feature alignment improves both intelligibility and matchability.

The SLM addresses prominent artifacts in alaryngeal speech such as aperiodicity and articulation blur that would otherwise propagate into the vocoding stage. By contextualizing symbolic patterns via masked token modeling, the SLM regularizes noisy embeddings into more canonical phonetic representations. This benefit is particularly pronounced in challenging conditions such as low signal-to-noise ratios or speech from non-native speakers.

2) Impact of Attention-Driven Feature Matcher

To isolate the effect of our attention-driven alignment mechanism, we replaced the traditional nearest-neighbor matching module in the kNN-VC pipeline while preserving the original encoder and vocoder. This configuration achieved a significant FAD reduction to 5.72, corresponding to a 23.8% improvement over the baseline. This demonstrates the effectiveness of our attention-based matcher, which replaces static cosine similarity with a soft-attention fusion mechanism over k -nearest neighbors in WavLM embedding space. Unlike conventional matching that selects a single best match, our method attends over multiple candidates using scaled dot-product attention and residual integration. This approach allows context-aware feature selection and reduces mismatches caused by alaryngeal distortions, leading to better phonetic and prosodic coherence in the converted output.

3) Full System Performance

The complete system combines both the SLM-based source enhancement and the Attention-Driven Feature Matcher. This configuration yields the best overall performance, achieving a FAD score of 4.83—representing a 35.7% reduction from the baseline, and outperforming the individual improvements of SLM (12.7%) and attention-based matching (23.8%).

The synergistic integration of both components delivers benefits greater than the sum of their parts. The enhanced input features improve matchability, while the attention-based matcher further refines alignment precision. Together, these modules form a robust pipeline that addresses both upstream (signal quality) and downstream (feature alignment) bottlenecks in alaryngeal speech enhancement.

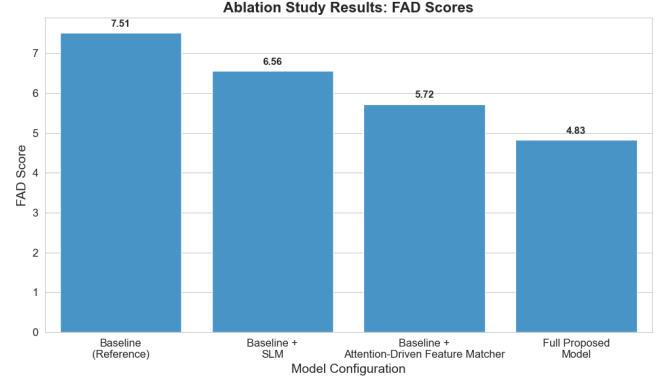


FIGURE 8. Component-wise Ablation Study Results (FAD scores). Lower is better.

B. Datasets

We constructed a multilingual dataset combining both Thai and English speech samples to evaluate the cross-linguistic generalizability of our framework. Given the limited availability of English alaryngeal speech, we synthesized English alaryngeal speech using our proposed MaskCycleGAN-VC model to simulate realistic pathological variability.

Feature representations were extracted using the 6th layer of the pretrained WavLM model, which provides robust phonetic and prosodic embeddings. The datasets used in our experiments are as follows:

- **Thai Speech:** 1,250 real alaryngeal samples from the dataset in [19] and 1,250 normal speech samples from the TSynC-2 corpus [20]. Figure 9 illustrates the train/validation/test split distribution.
- **English Speech:** 1,250 synthetic alaryngeal samples generated using our augmentation method [21], and 1,250 normal speech samples from the Mozilla Common Voice corpus [22]. The dataset distribution is shown in Figure 10.

C. Hyperparameter Configuration and Training Procedures

To ensure optimal performance and generalization, we carefully tuned hyperparameters across all modules of our proposed system. This section details the architectural specifications and training setups for each component, including pretraining strategies, loss configurations, and optimization schedules.

1) Speech Encoder

The speech encoder discretizes 1024-dimensional WavLM features into symbolic token sequences using K -means clustering with $K = 512$ centroids. Feature vectors are extracted at 20 ms intervals. To stabilize assignments and reduce quantization noise, batch-wise centroid averaging is

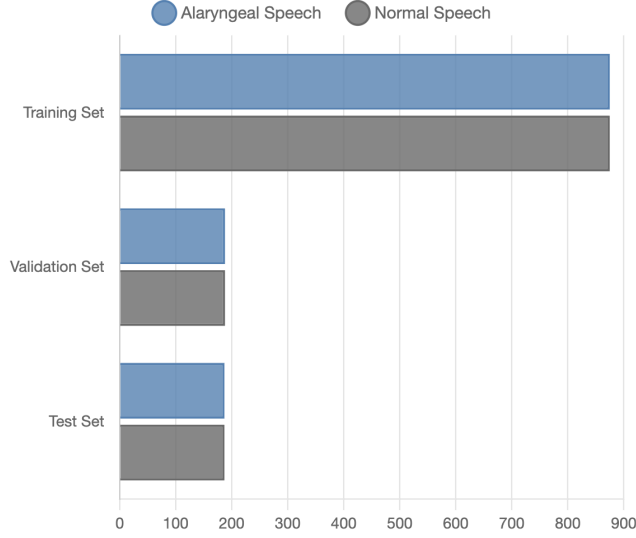


FIGURE 9. Thai speech data distribution for training, validation, and testing.

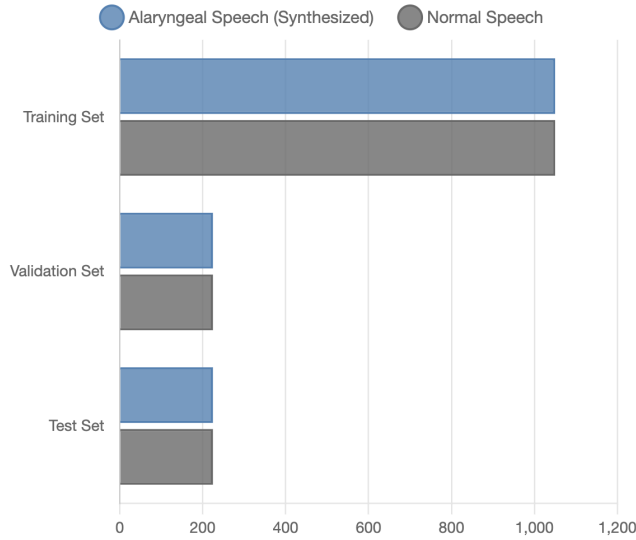


FIGURE 10. English speech data distribution for training, validation, and testing.

applied via efficient nearest-neighbor search. The encoder is pretrained jointly on both normal and alaryngeal speech to ensure robust token boundaries across speaking conditions.

2) Speech Language Model (SLM)

The SLM is implemented as a 6-layer causal Transformer with 8 attention heads and a hidden embedding size of 512. We employ GELU activation, sinusoidal positional encodings, and a dropout rate of 0.1. During inference, nucleus sampling with $p = 0.92$ is used to balance generation

diversity and fidelity. The model operates on discrete token sequences, refining them via masked span modeling to correct articulation inconsistencies and enforce phonotactic regularities.

3) Detokenizer

The detokenizer reconstructs frame-level acoustic features from symbolic tokens using a 4-layer Conformer decoder. Each layer comprises:

- Multi-head self-attention (8 heads, 256 dimensions),
- Depthwise separable convolutions (kernel size = 31),
- GLU-activated convolutional blocks with Swish nonlinearity.

The output is linearly projected back to the original WavLM feature space (1024 dimensions, 20 ms resolution). This module ensures temporally aligned and perceptually plausible reconstructions.

Training Procedure for SLM and Detokenizer:

Both components are jointly trained on Thai and English corpora using the masked language modeling objective [6]. The training configuration includes:

- Sequence length: 500 tokens,
- Batch size: 32,
- Learning rate: 1×10^{-4} with AdamW,
- Masking ratio: 15% tokens per batch,
- Epochs: 100.

4) Attention-Driven Feature Matcher

The matcher module consists of three subcomponents:

Feature Enhancer:

A bottleneck MLP with architecture:

$$1024 \rightarrow 256 \rightarrow 256 \rightarrow 1024$$

Dropout of 0.3 is applied after each layer. A learnable residual weighting parameter α is initialized to 0.5 and updated during training to balance enhanced and original embeddings.

Attention Module:

This module computes attention over the $K = 10$ nearest candidates using 4-headed scaled dot-product attention in the 1024-dimensional embedding space. It replaces conventional nearest-neighbor matching with a learned soft-alignment mechanism.

Quality Predictor:

A two-layer MLP that evaluates the perceptual quality of transformed features:

$$1024 \rightarrow 128 \rightarrow 1$$

The first layer uses ReLU activation, followed by a sigmoid output layer to produce quality scores $\hat{q} \in [0, 1]$. This component provides additional supervisory signal to guide feature transformation toward natural speech characteristics.

Training Configuration:

- Optimizer: AdamW,
- Learning rate: 1×10^{-4} , weight decay: 1×10^{-4} ,
- Scheduler: Cosine annealing over 50 epochs,
- Batch size: 32.

Loss Composition:

The multi-objective training combines the following components:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}} + \lambda_{\text{con}}\mathcal{L}_{\text{con}} + \lambda_{\text{qual}}\mathcal{L}_{\text{qual}}$$

with weights:

- $\lambda_{\text{rec}} = 1.5$,
- $\lambda_{\text{reg}} = 0.05$,
- $\lambda_{\text{con}} = 0.05$,
- $\lambda_{\text{qual}} = 0.1$.

5) MaskCycleGAN-VC for Data Augmentation

To expand the diversity of training samples and simulate real-world alaryngeal distortions, we employed MaskCycleGAN-VC with structured spectrogram masking (Section A). The model uses:

- A 2-1-2D CNN-based generator,
- A PatchGAN discriminator [23],
- Audio downsampling to 22.05 kHz,
- 80-bin log-Mel spectrograms computed with a 1024-sample window and 256-sample hop.

Training Setup:

- Optimizer: Adam,
- Learning rates: 0.0002 (generator), 0.0001 (discriminator),
- Momentum: $\beta_1 = 0.5$, $\beta_2 = 0.999$,
- Batch size: 1, processing 64-frame crops (~ 0.75 seconds),
- Iterations: 5,000,
- Validation split: 10% of training data,
- Loss weights: $\lambda_{\text{cyc}} = 10$, $\lambda_{\text{id}} = 5$, $\lambda_{\text{te}} = 0.001$.

This augmentation setup enabled us to simulate diverse and pathological speaking conditions that enhance generalization under clinical constraints.

D. Objective Evaluation

We conducted a comprehensive objective evaluation to assess the effectiveness of our proposed alaryngeal speech enhancement system across both Thai and English datasets. The evaluation comprises 50 test samples from 3 different alaryngeal speakers per language, ensuring phonetic diversity

TABLE 2. Objective intelligibility and speaker similarity for Thai alaryngeal speech.

Model	FAD ↓	MCD ↓	SIM ↑	MOS ↑
MaskCycleGAN-VC [8]	12.057	17.170	0.696	2.407
kNN-VC [3]	8.888	7.855	0.790	2.710
Wesper [24]	7.696	8.284	0.620	3.371
Phoneme Hallucinator [5]	12.698	11.641	0.789	2.231
SELM [6]	13.896	13.162	0.565	0.829
MUSE [25]	10.926	8.511	0.530	2.004
Proposed Method	5.940	6.257	0.820	3.401

while maintaining controlled speaker conditions. Tables 2 and 3 report performance in terms of Fréchet Audio Distance (FAD) [26], Mel-Cepstral Distortion (MCD) [27], speaker similarity (SIM), and mean opinion scores (MOS).

Our model consistently outperforms prior state-of-the-art methods across all evaluation criteria. Specifically, we achieve the lowest FAD and MCD scores—5.940 and 6.257 for Thai, and 4.826 and 5.590 for English, respectively—indicating significant improvements in both perceptual and spectral fidelity. The proposed method also demonstrates superior speaker preservation, achieving SIM scores of 0.820 (Thai) and 0.784 (English), outperforming competitive systems such as kNN-VC, Wesper, and Phoneme Hallucinator.

Relative to our primary baseline, kNN-VC, our approach yields a 33% reduction in FAD and 20% reduction in MCD for Thai, and a 36% and 8% reduction, respectively, for English. These gains validate the effectiveness of our dual-module strategy—source enhancement via the Speech Language Model and feature alignment via the Attention-Driven Feature Matcher.

Furthermore, our method achieves the highest MOS score of 4.511 for English, and strong performance for Thai, indicating perceptual preference among listeners. The controlled evaluation across multiple speakers and languages demonstrates the robustness of our approach, positioning our system as a new state-of-the-art benchmark for alaryngeal voice conversion across diverse phonetic and linguistic conditions.

Additionally, Table 4 provides a comparison of objective metrics for different masking strategies employed in alaryngeal speech data generation.

E. Subjective Evaluation

To complement the objective metrics, we conducted a subjective evaluation via ABX testing to assess the perceptual impact of our proposed enhancement method. The test involved 20 participants: 10 native Thai speakers and 10 native English speakers, all with self-reported normal hearing and familiarity with synthetic speech. Evaluations were performed in acoustically controlled environments to minimize external bias.

TABLE 3. Objective intelligibility and speaker similarity for English alaryngeal speech.

Model	FAD ↓	MCD ↓	SIM ↑	MOS ↑
MaskCycleGAN-VC [8]	19.331	5.278	0.558	2.686
kNN-VC [3]	7.518	6.086	0.626	3.397
Wesper [24]	5.581	5.685	0.647	4.422
Phoneme Hallucinator [5]	6.503	5.662	0.737	4.085
SELM [6]	9.195	5.646	0.705	3.567
MUSE [25]	13.270	6.654	0.549	3.858
Proposed Method	4.826	5.590	0.784	4.511

The ABX test setup was designed to assess listener discrimination between (A) enhanced alaryngeal speech, (B) original unprocessed alaryngeal speech, and (X) normal speech. Each trial required listeners to determine whether sample X was perceptually closer to A or B. This framework offers a reliable measure of relative perceptual quality and similarity to natural speech.

To evaluate rating consistency across participants, we computed pairwise Pearson correlation coefficients among five randomly selected raters from each language group. Results are reported in Tables 5 (English) and 6 (Thai).

For English speech, strong agreement was observed among several rater pairs—particularly R3 vs. R4 ($r = 0.871$, $p = 0.001$), R3 vs. R5 ($r = 0.722$, $p = 0.018$), and R4 vs. R5 ($r = 0.829$, $p = 0.003$)—indicating a high degree of inter-rater reliability. However, correlations involving R1 and R2 were moderate (e.g., R1 vs. R2: $r = 0.509$), suggesting potential subjectivity or perceptual variation in their judgments.

In contrast, the Thai rater group demonstrated more uniformly high inter-rater agreement. Notably, R1 vs. R4 achieved a strong correlation of $r = 0.885$ ($p = 0.001$), with similarly high values observed for R2 vs. R3 ($r = 0.722$), R3 vs. R4 ($r = 0.667$), and R1 vs. R5 ($r = 0.645$), all of which were statistically significant. These results affirm the consistency and reliability of subjective evaluations among Thai listeners.

A cross-linguistic comparison revealed that participants consistently preferred the enhanced speech over the unprocessed counterpart. However, selection rates were notably higher among Thai listeners. We attribute this to differences in training data: the Thai enhancement model was trained on real clinical alaryngeal speech, whereas the English model relied on synthetically generated samples. Real-world alaryngeal speech contains complex articulatory patterns, dynamic vocal effort, and speaker-specific nuances that synthetic data may fail to replicate fully. This mismatch likely accounts for the greater variability in perceptual evaluations among English-speaking participants.

These findings highlight the value of training with authentic pathological speech data to ensure the generalizability and

effectiveness of enhancement models in real-world deployment scenarios.

VI. Conclusion

This work presents a comprehensive framework for alaryngeal speech enhancement, combining several state-of-the-art components to significantly improve intelligibility, naturalness, and speaker identity preservation for individuals post-laryngectomy. Central to our approach is a phoneme refinement pipeline driven by a Speech Language Model (SLM) trained on discrete representations derived from SELM. These refined representations facilitate improved alignment with canonical speech patterns via an Attention-Driven Feature Matcher, which enhances spectral-temporal consistency and leads to a substantial 43.5% reduction in Fréchet Audio Distance compared to kNN-VC baselines.

To address data scarcity, we further introduce a novel augmentation strategy using MaskCycleGAN-VC equipped with a timbre similarity loss. This ensures that synthetic alaryngeal speech not only diversifies training data but also retains speaker-specific timbral properties critical for personalized assistive applications.

Extensive evaluations—including both objective metrics and subjective ABX testing across Thai and English speakers—demonstrate the superiority of the proposed system over competitive baselines. Strong inter-rater reliability (Pearson $r = 0.327$ – 0.885) supports the validity of listener judgments, with consistently higher ratings for enhanced samples. Importantly, our method maintains speaker identity while improving intelligibility across a range of prosodic and phonetic conditions.

Future directions include integration of adaptive language modeling to improve speaker-dependent generalization, exploration of larger multilingual corpora, and extension to generative architectures such as diffusion models [28] for finer speech synthesis control. Overall, our contributions lay important groundwork for advancing personalized speech restoration technologies and enhancing the quality of life for individuals with vocal impairments.

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TABLE 4. Objective evaluation across masking strategies for alaryngeal speech generation.

Masking Method	MCD		FAD		Timbre Similarity	
	Synth vs Real	Real vs Real	Synth vs Real	Real vs Real	Synth vs Real	Real vs Real
Consecutive Time Masking (CTM)	3.7	3.0	6.5	3.0	0.78	0.80
Articulatory Dimension Masking (ADM)	3.5	3.0	7.1	3.0	0.72	0.80
SpecAugment [13]	3.6	3.0	7.2	3.0	0.75	0.80

TABLE 5. Inter-rater Pearson Correlation Coefficients for ABX Test (English Speech)

	R1 vs. R2	R1 vs. R3	R1 vs. R4	R1 vs. R5	R2 vs. R3	R2 vs. R4	R2 vs. R5	R3 vs. R4	R3 vs. R5	R4 vs. R5
Pearson r	0.509	0.509	0.395	0.327	0.259	0.201	0.167	0.871	0.722	0.829
p -value (2-tailed)	0.133	0.133	0.259	0.356	0.469	0.578	0.645	0.001	0.018	0.003
N	10	10	10	10	10	10	10	10	10	10

TABLE 6. Inter-rater Pearson Correlation Coefficients for ABX Test (Thai Speech)

	R1 vs. R2	R1 vs. R3	R1 vs. R4	R1 vs. R5	R2 vs. R3	R2 vs. R4	R2 vs. R5	R3 vs. R4	R3 vs. R5	R4 vs. R5
Pearson r	0.330	0.547	0.885	0.645	0.722	0.481	0.561	0.667	0.401	0.459
p -value (2-tailed)	0.352	0.102	0.001	0.044	0.018	0.159	0.092	0.035	0.250	0.182
N	10	10	10	10	10	10	10	10	10	10

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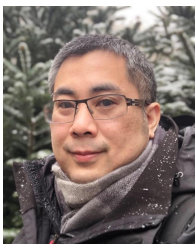
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